

Hector, 55 · 6 months of **epigastric burning**, worse at night, relieved by food · weight loss · today: **black tarry stool (melena)** · stool antigen positive for **H. pylori** · endoscopy: **bleeding duodenal ulcer**

BIO 004 · WEEK 5 · CASE DEEP DIVE

The Alimentary Canal

A patient who lost weight without trying, a black stool, and a bacterium that has lived in his stomach for decades.

PATIENT CASE

Hector is 55, a construction foreman. Over the past six months he has had **epigastric burning** that wakes him at 2 AM and is briefly relieved when he eats. He has lost 10 pounds without trying. This morning he passed a **black, tarry stool** and felt lightheaded.

In the ED his hemoglobin is 9.5 (down from a year ago). Stool antigen test is **positive for Helicobacter pylori**. Upper endoscopy reveals a **1-cm ulcer in the duodenal bulb with a visible vessel**, which the gastroenterologist clips. He is started on a proton pump inhibitor and triple-therapy antibiotics for H. pylori.

The alimentary canal is a continuous tube from mouth to anus, but its wall changes character at every region to do a different job. The case rests on understanding what the wall of the stomach and duodenum look like, what protects them from acid, and what happens when that protection fails.

GUIDING QUESTION

What are the regions of the alimentary canal, what are the four layers of the gut wall, and how does the lining of the stomach protect itself from acid that would dissolve any other tissue?

PREWORK

Companion to your Week 5 Alimentary Canal prework page. Terms, video, and spaced recall for the oral cavity, pharynx, esophagus, stomach, small intestine, large intestine, and the four-layer wall live there.

1 PLAN

One tube, ~9 meters long, with regional specialties

The alimentary canal runs from mouth to anus. Each region modifies the basic structure to do a specific job: mouth (mechanical breakdown, salivary digestion), pharynx and esophagus (transport), stomach (mixing, acid digestion of protein), small intestine (most digestion and absorption), large intestine (water reclamation, microbial fermentation, stool formation).

Mouth

Teeth, tongue, salivary glands.

Pharynx + esophagus

Conduit, 25 cm. Skeletal muscle upper third, smooth muscle lower third.

Stomach

J-shaped reservoir; acid + pepsin. Where Hector's pain starts.

Small intestine

~6 m: duodenum, jejunum, ileum. Most digestion and absorption.

Large intestine

~1.5 m: cecum, colon (asc/trans/desc/sigmoid), rectum, anal canal.

Four layers, the same recipe with regional variation

● Mucosa

Innermost. Epithelium + lamina propria (CT) + muscularis mucosae. Type of epithelium varies by region.

● Submucosa

Dense CT with blood vessels, lymphatics, and the *submucosal (Meissner's) plexus* of the enteric nervous system.

● Muscularis externa

Inner circular + outer longitudinal smooth muscle. *Myenteric (Auerbach's) plexus* between them drives peristalsis.

● Serosa

Visceral peritoneum on intraperitoneal organs; *adventitia* (no serosa) where the tube is retroperitoneal (esophagus, much of duodenum, ascending and descending colon).

Four regions, three cell types, one defense problem

Four anatomical regions: **cardia** (entrance), **fundus** (dome), **body** (largest portion), **pylorus** (exit through the pyloric sphincter). The wall has an extra oblique muscle layer (third layer beyond inner circular + outer longitudinal) for vigorous mixing.

Gastric pits open into **gastric glands**, which contain:

● **Parietal cells**

Secrete

HCl

(pH ~1-2) and intrinsic factor (needed for B12 absorption).

● **Chief cells**

Secrete pepsinogen, activated to pepsin in acid.

● **Mucous neck cells + surface mucous cells**

Secrete bicarbonate-rich mucus that coats the lining, protecting it from acid.

● **G cells**

In the pyloric antrum. Secrete *gastrin*, which stimulates parietal cells.

Why peptic ulcers form. The stomach's defense is the mucus/bicarbonate layer plus tight junctions and rapid epithelial turnover. *H. pylori* colonizes the mucus layer, secretes urease (raises local pH), and triggers chronic inflammation that disrupts the mucosal barrier. *NSAIDs* inhibit prostaglandins that maintain mucus production. Either mechanism lets acid digest the underlying tissue, producing an ulcer. Duodenal ulcers (Hector's) are most often *H. pylori* related; gastric ulcers are split more evenly between *H. pylori* and *NSAIDs*.

Three regions, all designed to maximize surface area

● Duodenum (25 cm)

C-shaped, wraps around head of pancreas.

Receives chyme from stomach, bile from gallbladder, enzymes from pancreas. *Hector's ulcer site (duodenal bulb).*

● Jejunum (~2.5 m)

Tallest plicae circulares, longest villi, most absorption.

● Ileum (~3.5 m)

Peyer's patches (immune surveillance). Absorbs B12 (with intrinsic factor) and bile salts. Ends at ileocecal valve.

Three levels of surface-area amplification: **plicae circulares** (large mucosal folds, visible to the eye) → **villi** (finger-like projections of mucosa, visible with low magnification) → **microvilli** (brush border on enterocyte apical surface, visible only with EM). Total absorptive surface is ~250 m² (a tennis court).

Reclamation, fermentation, formation

● Cecum + appendix

Cecum: blind pouch. Appendix: lymphoid tissue, can obstruct → appendicitis.

● Colon (asc/trans/desc/sigmoid)

Three distinctive features: *teniae coli* (three longitudinal bands), *haustra* (pouches), *epiploic appendages* (fat tags).

● Rectum

Storage. No haustra. Transverse rectal folds slow descent.

● Anal canal

Internal anal sphincter (smooth, involuntary) + external (skeletal, voluntary).

Mucosa lacks villi (no significant absorption beyond water and electrolytes). Crypts have abundant goblet cells producing mucus to lubricate progressively dehydrating stool.

Three branches from the abdominal aorta

Celiac trunk: stomach, liver, spleen, proximal duodenum, pancreas. **Superior mesenteric artery (SMA):** distal duodenum, jejunum, ileum, ascending and proximal transverse colon.

Inferior mesenteric artery (IMA): distal transverse, descending, sigmoid colon, upper rectum. **Venous drainage** from the GI tract enters the *hepatic portal system*: blood from gut capillaries → mesenteric/splenic veins → portal vein → liver sinusoids → hepatic veins → IVC. This is how absorbed nutrients reach the liver before the rest of the body.



DRAW ZONE 1 -- Sketch the four layers of the gut wall and label each. Then draw a normal stomach lining vs an ulcerated one with *H. pylori* in the mucus.

Fine pencil for the surface mucous layer. Show the ulcer crater extending through mucosa into submucosa.



DRAW ZONE 2 -- Draw the alimentary canal from mouth to anus. Label each region and the three blood-supply territories (celiac, SMA, IMA).

Color the celiac/SMA/IMA territories differently. Mark Hector's ulcer in the duodenal bulb (celiac territory).

RETURNING TO HECTOR

An ulcer, a bacterium, and a wall that failed

Hector's case is the gut wall under attack from inside.

"Why pain at 2 AM, relieved by food"

Duodenal ulcer pain classically wakes patients in the early morning when stomach acid pours into an empty duodenum. Food (and antacids) buffer the acid and bring temporary relief.

"Black tarry stool (melena)"

Upper GI blood is digested as it transits → black, tarry, foul. Confirms upper GI bleeding (gastric or duodenal). Bright red blood from the rectum (hematochezia) suggests lower GI source.

"Weight loss"

Pain → food avoidance. Also chronic blood loss → iron deficiency anemia → fatigue, reduced appetite.

"Why H. pylori is the smoking gun"

It lives in the mucus layer, raises local pH with urease, triggers chronic gastritis, and undermines the mucosal barrier. 90% of duodenal ulcers and 70% of gastric ulcers are H. pylori related.

"Visible vessel at the ulcer base"

Erosion into submucosal artery. High rebleed risk. Endoscopic clipping, thermal coagulation, or injection therapy reduces risk.

"PPI + triple therapy"

PPI heals the ulcer by reducing acid; antibiotics (typically amoxicillin + clarithromycin + PPI) eradicate H. pylori. Confirm eradication after treatment with stool antigen or breath test.

Mucosa, mucus, acid, bacterium. Hector's ulcer is the wall of the gut failing in slow motion under chronic insult.

